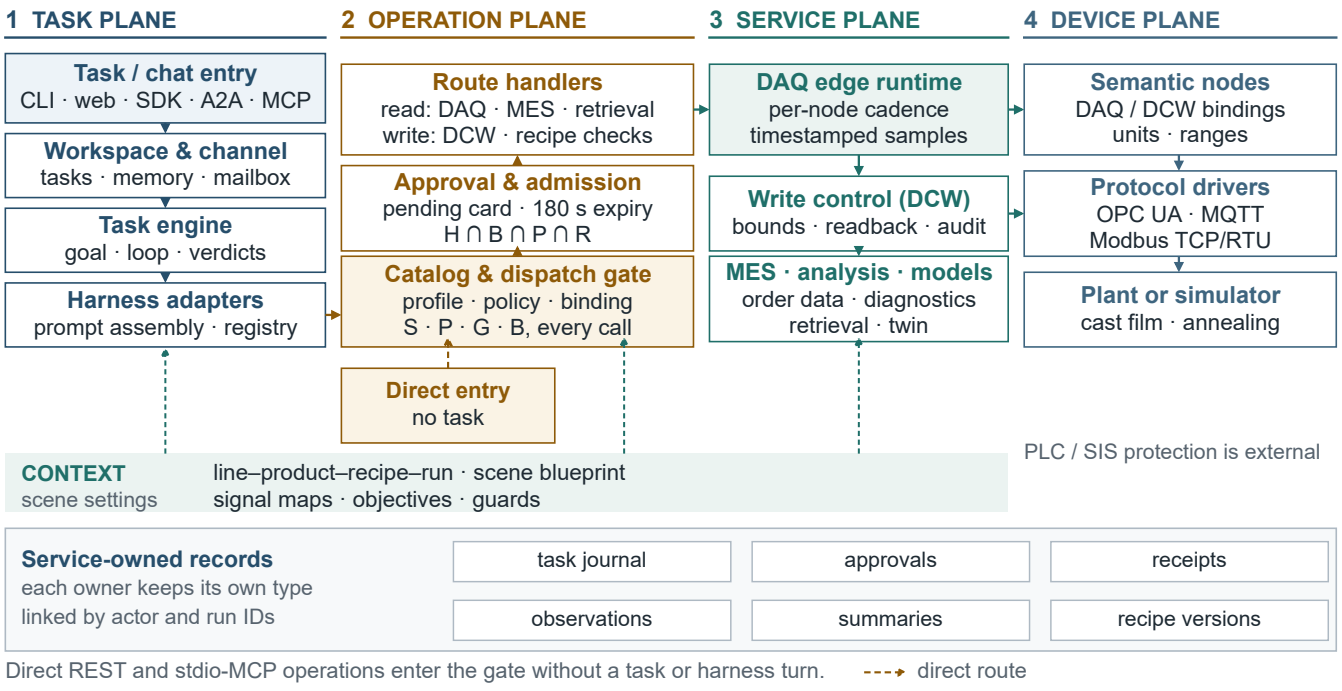

1 TASK PLANE
2 OPERATION PLANE
3 SERVICE PLANE
4 DEVICE PLANE
Task / chat entry
CLI · web · SDK · A2A · MCP
Workspace & channel
tasks · memory · mailbox
Task engine
goal · loop · verdicts
Harness adapters
prompt assembly · registry
Route handlers
read: DAQ · MES · retrieval
write: DCW · recipe checks
Approval & admission
pending card · 180 s expiry
H ∩ B ∩ P ∩ R
Catalog & dispatch gate
profile · policy · binding
S · P · G · B, every call
Direct entry
no task
DAQ edge runtime
per-node cadence
timestamped samples
Write control (DCW)
bounds · readback · audit
MES · analysis · models
order data · diagnostics
retrieval · twin
Semantic nodes
DAQ / DCW bindings
units · ranges
Protocol drivers
OPC UA · MQTT
Modbus TCP/RTU
Plant or simulator
cast film · annealing
PLC / SIS protection is external
CONTEXT
scene settings
line–product–recipe–run · scene blueprint
signal maps · objectives · guards
Service-owned records
each owner keeps its own type
linked by actor and run IDs
task journal
approvals
receipts
observations
summaries
recipe versions
Direct REST and stdio-MCP operations enter the gate without a task or harness turn.
direct route